11 Agent planning and feedback

This chapter covers

- Planning for an LLM and implementing it in agents and assistants
- Using the OpenAI Assistants platform via custom actions
- Implementing/testing a generic planner on LLMs
- Using the feedback mechanism in advanced models
- Planning, reasoning, evaluation, and feedback in building agentic systems

Now that we've examined how large language models (LLMs) can reason and plan, this chapter takes this concept a step further by employing planning within an agent framework. Planning should be at the core of any agent/assistant platform or toolkit. We'll start by looking at the basics of planning and how to implement a planner through prompting. Then, we'll see how planning operates using the OpenAI Assistants platform, which automatically incorporates planning. From there, we'll build and implement a general planner for LLMs.

Planning can only go so far, and an often-unrecognized element is feedback. Therefore, in the last sections of the chapter, we explore feedback and implement it within a planner. You must be familiar with the content of chapter 10, so please review it if you need to, and when you're ready, let's begin planning.

11.1 Planning: The essential tool for all agents/assistants

Agents and assistants who can't plan and only follow simple interactions are nothing more than chatbots. As we've seen throughout this book, our goal isn't to build bots but rather to build autonomous thinking agents—agents that can take a goal, work out how to solve it, and then return with the results.